Every
Ware
Lab

UNIVERSITÀ
DEGLI STUDI
DI MILANO

eXplainable AI for Ambient Intelligence

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

Gabriele Civitarese

EveryWare Lab, Department of Computer Science

Università degli Studi di Milano

Copyright

Every Ware Lab

Intro to eXplainable AI

Every
Ware
Lab

Machine Learning in Ambient Intelligence

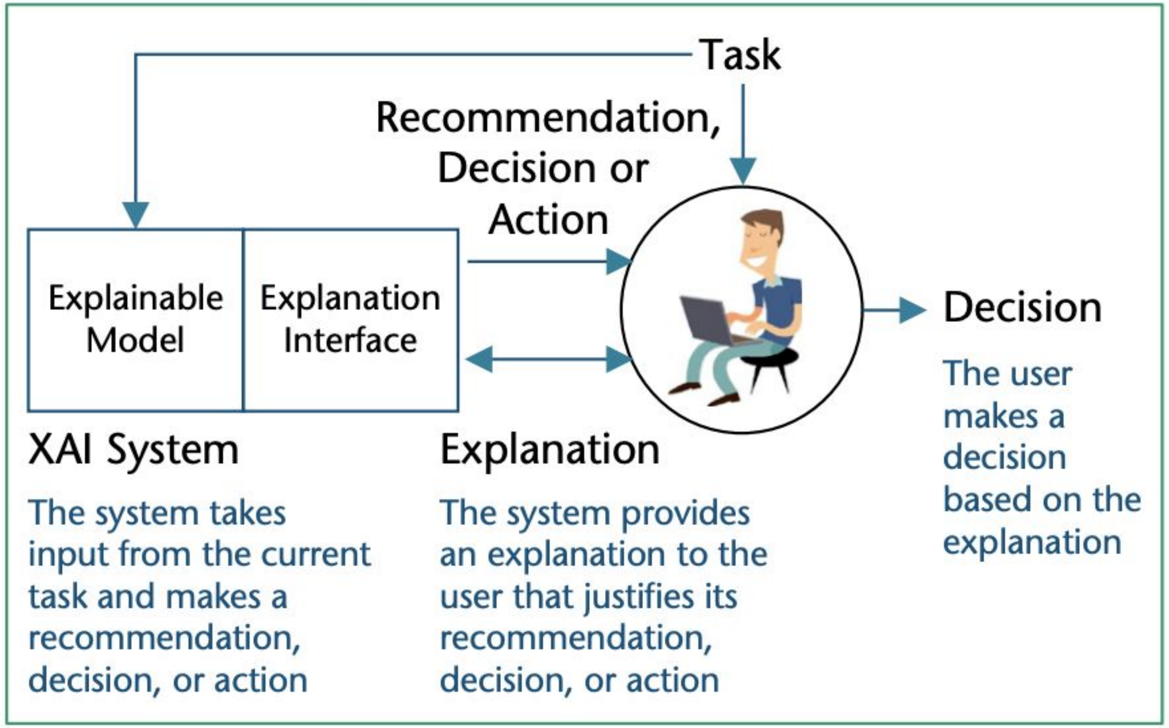
- As we learned in this course, machine learning models play a crucial role in Ambient Intelligence scenarios
- Their output is often crucial for important decisions in several domains, for instance:
  - supporting clinicians in healthcare diagnosis
  - supporting management decisions in smart cities
  - supporting smart-home energy saving choices
  - supporting data scientists in improving the model or the sensing setting
  - …

Every Ware Lab

eXplainable AI (XAI)

- In some domains, it is not only important to know ***what*** it is predicted, but also ***why***
- Knowing the ‘why’ helps in learning more about the problem, the data and the reason why a model might fail
- Explainable AI (XAI) techniques unveil the reasoning behind the predictions and decisions of machine learning models
- XAI gained a lot of attention in Computer Vision and NLP domains, but it is also crucial for time-series sensor data in several fields of Ambient Intelligence

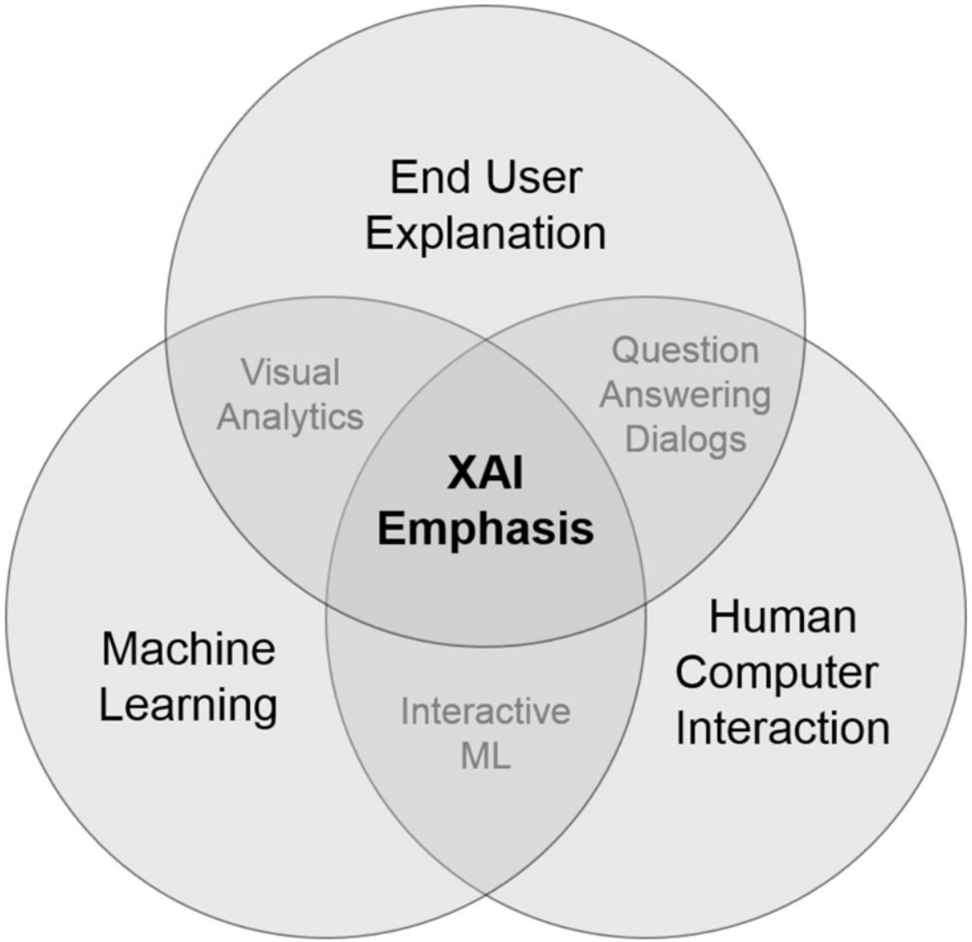
A typical Explanation Framework



Every Ware Lab

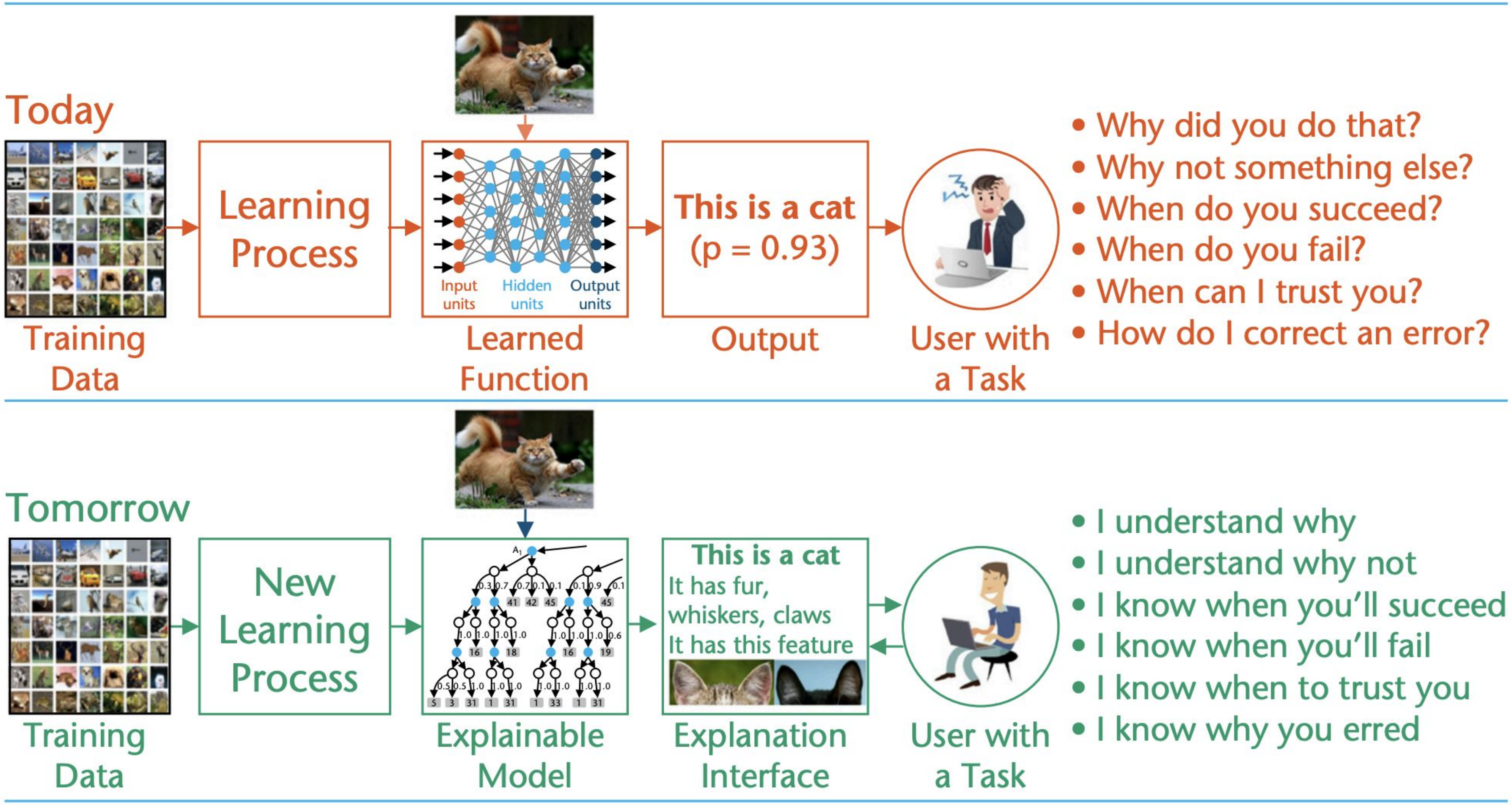
XAI Emphasis

- XAI is not only about machine learning models
- Human Computer Interaction (HCI) plays a major role
  - explanations should be also *effectively* delivered to target users!
- There is also a strong “psychological” to consider when designing how to deliver explanations



Every Ware Lab

XAI: the vision



Every Ware Lab

The target of explanations

- When building XAI models, it is crucial to determine the ***target users*** that will use the explanations
- There are indeed two main categories:
  - ***non-expert end-users***: they take advantage of AI models in their daily life without knowledge about machine learning
    - e.g., in AmI scenarios they may be smart city operators, clinicians, monitored subjects
  - ***technicians:*** data scientists/machine learning experts
    - it is crucial for them to monitor the model to understand if the sensing setup requires refinements, if more labeled data need to be collected, if the model has to be improved, etcetera
- Based on the target, explanations may differ a lot!

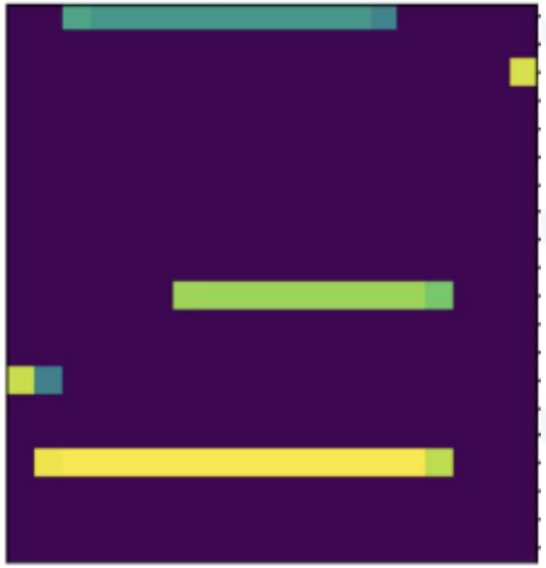
Every Ware Lab

Examples of explanations: end-user vs technician

Explanation for non-experts

Patrick was Watching the Television mainly because he has just Eaten, and he has been in the Living Room. I also noticed that he was sitting on the Living Room Couch and that the Television was on.

Explanation for Technicians



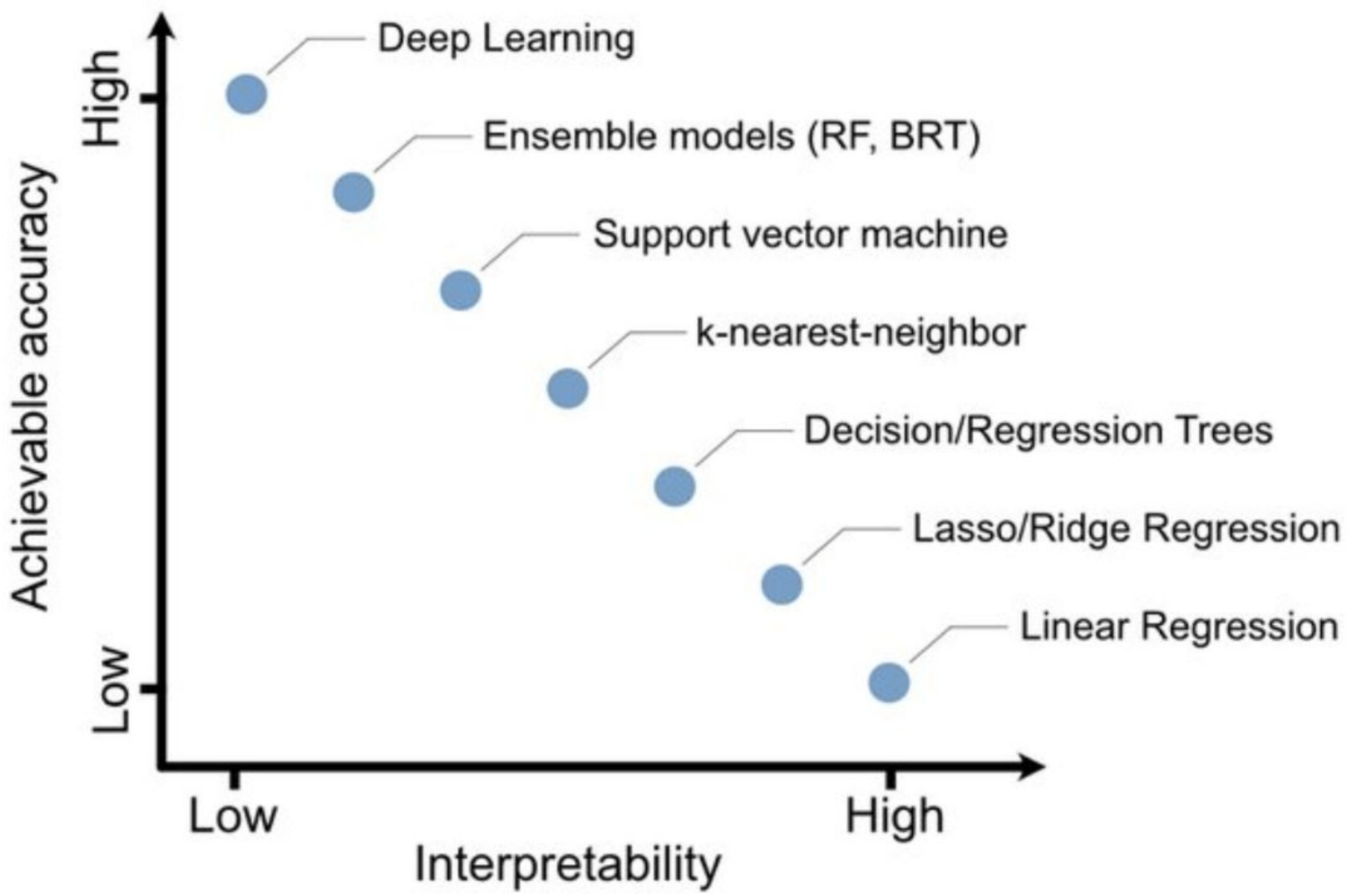
Global VS Local Explanations

- ***Global Explainability:***
  - *How does the trained model make predictions?*
  - *Which features are important in general and what kind of interactions between them take place?*
  - Usually, it is prohibitive to understand the *whole* model, but sometimes it is possible to explain a portion of it (e.g., specific weights)

- ***Local Explainability***
  - *Why did the model make a certain prediction for an instance?*
  - Locally, the prediction might depend only on some features
  - Most widely adopted since it is more accurate than making global explanations

Every Ware Lab

Accuracy VS Interpretability



Every
Ware
Lab

Main categories of XAI approaches

- In this lecture, we will explore three main categories of XAI methods:
  - Interpretable Models
  - Model Agnostic
  - Deep Explanation

Interpretable Models

Interpretable XAI models

- Some classic ML algorithms are inherently explainable
- Once they are trained, their parameters may be used to deliver explanations
  - global or local, depending on the specific model
- **Pro:** very simple approach!
- **Cons:** for certain tasks, classic ML models are too simple compared to the ones based on deep learning
- In the following we will see only a few of them (the most representative ones)

Every Ware Lab

Linear Regression

- A linear regression model makes a prediction with a weighted sum of the feature inputs
- The weights $\beta_1,\ldots,\beta_p$ indicate the feature importance and hence can be used as a global explanation
- **Pros**:
  - simple weights estimation procedure
  - it is easy to understand the interpretation (i.e., high weights are associated with most important features)
- **Cons**:
  - linearity does not allow to capture interaction between features
  - low predictive performances

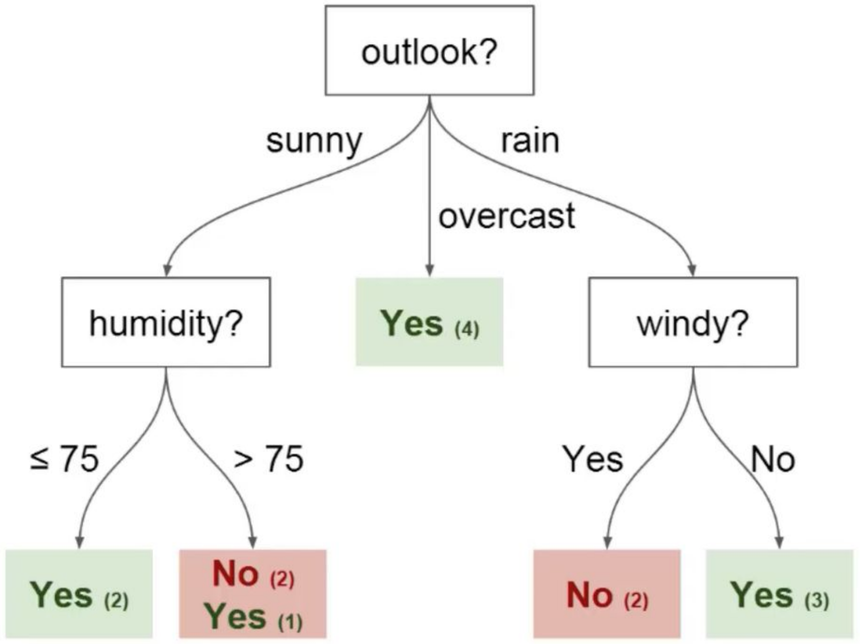
$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p + \epsilon$$

$$\hat{\beta} = \arg\min_{\beta_0,\ldots,\beta_p} \sum_{i=1}^{n} \left(y^{(i)} - \left(\beta_0 + \sum_{j=1}^{p} \beta_j x_j^{(i)} \right) \right)^2$$

Every Ware Lab

Decision Trees

- Decision trees can capture complex relationships among features
- It is possible to generate both global and local explanations once the decision tree is trained:
  - **global:** computing feature importance, by measuring at each node how much it has reduced the variance compared to the parent node
  - **local:** tracking the path of the data point from the root to the leaf, the combination of the traversed conditions is the local explanation



Every Ware Lab

Rules Induction

- Learning IF-THEN rules from data
- Each rule should cover a portion of the dataset
- There are several approaches that can be used
  - e.g., iterative approaches, pattern mining
- The set of rules used for prediction are the local explanation
- All the rules together compose a global explanation
- Examples of approaches: ***Sequential Covering, Bayesian Rules Lists***

if	bruises=no, odor=not-in-(none,foul)	then probability that the mushroom is edible	= 0.00112
else if	odor=foul, gill-attachment=free,	then probability that the mushroom is edible	= 0.0007
else if	gill-size=broad, ring-number=one,	then probability that the mushroom is edible	= 0.999
else if	stalk-root=unknown, stalk-surface-above-ring=smooth,	then probability that the mushroom is edible	= 0.996
else if	stalk-root=unknown, ring-number=one,	then probability that the mushroom is edible	= 0.0385
else if	bruises=foul, veil-color=white,	then probability that the mushroom is edible	= 0.995
else if	stalk-shape=tapering, ring-number=one,	then probability that the mushroom is edible	= 0.986
else if	habitat=paths,	then probability that the mushroom is edible	= 0.958
else	(default rule)	then probability that the mushroom is edible	= 0.001

Every Ware Lab

Model Agnostic Approaches

Every
Ware
Lab

Model Agnostic XAI

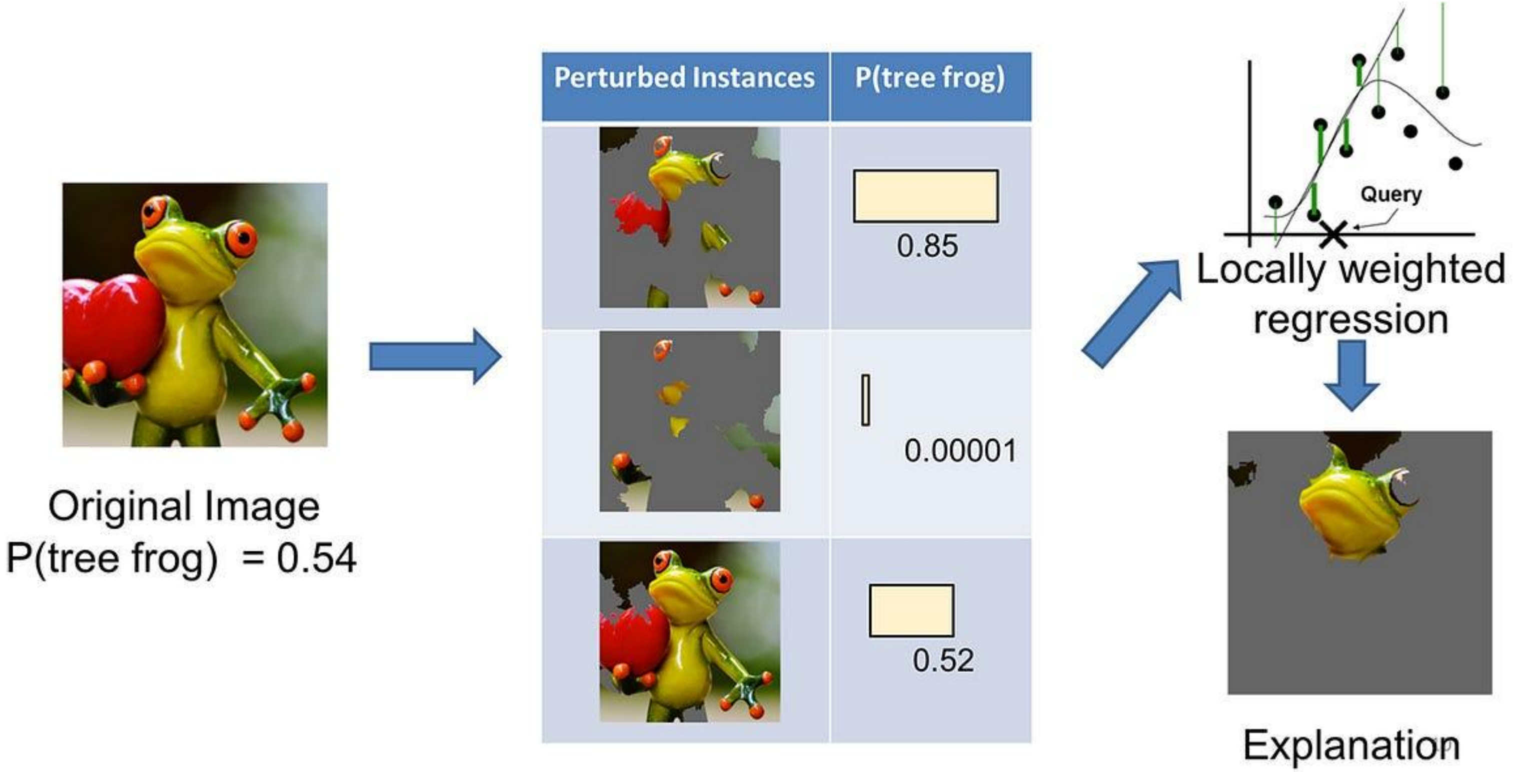
- Interpretable models are sometimes too simple and can not cope with complex domains
  - e.g., computer vision, natural language processing, time series analysis
- Model Agnostic XAI methods decouple explanations from the machine learning model
- Hence, a Model Agnostic XAI approach can be applied to any type of model (including deep learning) that is considered as a ***black box***
- Advantages:
  - **Model flexibility:** The interpretation method can work with any machine learning model (e.g., random forests, deep neural networks)
  - **Explanation flexibility:** It is possible to choose the best interpretation model based on the goal

Local Interpretable Model-agnostic Explanations (LIME)

- LIME analyzes how the model's output changes by varying the input
  - ***goal:*** finding the parts of the input are important for classification!
- Given a data sample which we want to explain:
  - generate a dataset of perturbed versions of the sample and the corresponding predictions using the black box model
  - use this training set to learn an inherently interpretable model (e.g., linear regression)
    - each perturbed sample should be weighted based on the similarity with the original data point!
  - the resulting global model explanation should accurately approximate the *local* prediction (local fidelity)

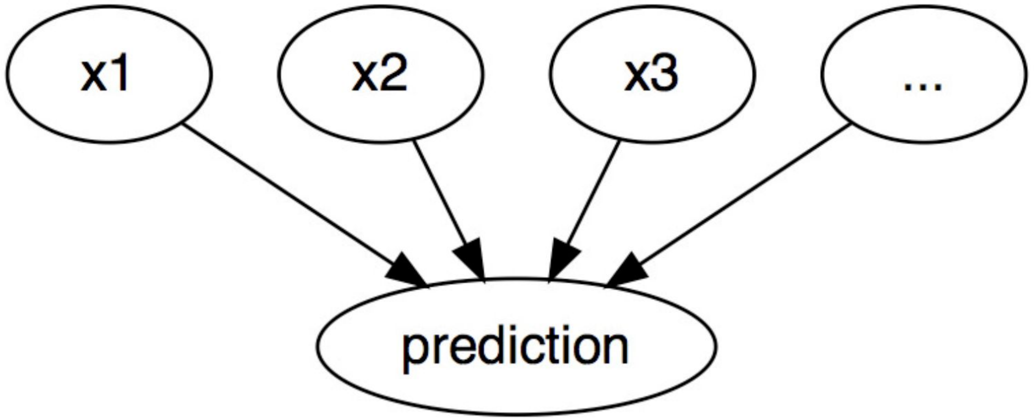
Every Ware Lab

LIME mechanism



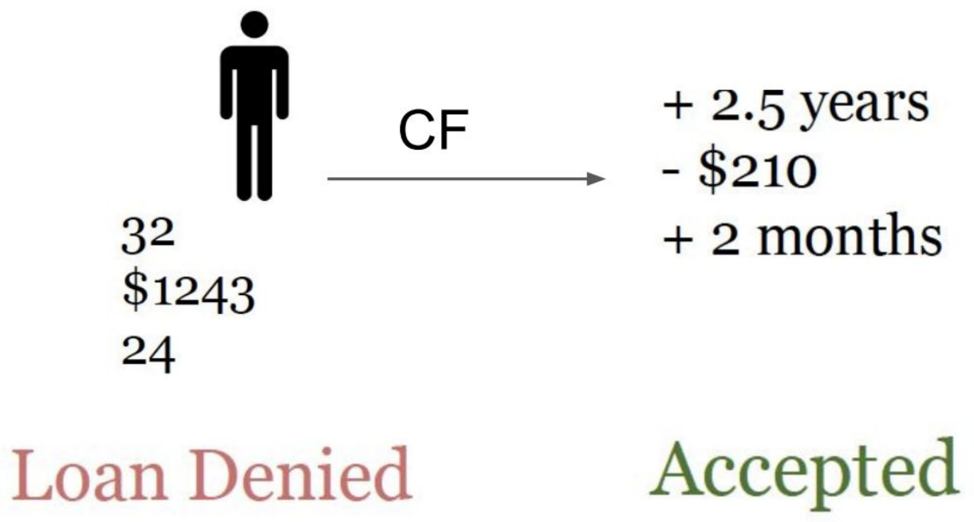
Counterfactual Explanations

- Describing a causal situation in the form: “If X had not occurred, Y would not have occurred”
  - X is the cause, Y is the event
  - imagining a hypothetical reality that contradicts the observed facts
  - e.g., “If I hadn’t taken a sip of this hot coffee, I wouldn’t have burned my tongue”
- Idea:
  - changing feature values (in a realistic way)
  - analyze how the prediction changes
  - observe when prediction changes in a relevant way
    - e.g., flipped prediction class
  - counterfactual explanation: a feature vector with the smallest change modifying the prediction output



Example of counterfactual explanations

- A 32 year-old male who wants to get a loan of $1243 for a duration of 24 months
- Counterfactual explanation:
    - *If he had he been 2.5 years older and requested $210 less for a duration two months shorter, he would have been eligible for the loan.*



Counterfactual Explanations: pros and cons

Pros:

- clear explanations
- easy to implement

Cons:

- For each instance it is possible to find multiple counterfactual explanations
- Multiple explanations may be contradictory, choosing the best may be challenging

Every
Ware
Lab

Problems of model-agnostic approaches

- While model-agnostic approaches are simple and flexible, they have some drawbacks
- Explanations do not always mimic the actual calculations made by the original model
- Indeed, the explanation may include different features compared to the ones actually used by the black box
- For this reason, methods to explain complex models have been proposed
    - e.g., deep learning models

Neural Network Interpretation

Every
Ware
Lab

Interpreting Neural Networks

- Methods to visualize features and concepts learned by a neural network, to explain individual predictions (local explanations)
- In deep learning models, a single prediction can involve millions of mathematical operations depending on the architecture of the neural network.
  - Often the number of weights is incredibly high
  - Impossible for humans to follow the exact mapping from data input to prediction!
  - Specific interpretation methods are needed
- Idea:
  - neural networks learn features and concepts in their hidden layers
  - the gradient can be utilized to implement interpretation methods
    - more computationally efficient than model-agnostic methods looking at the model "from the outside"

Gradient-weighted Class Activation Map (GradCAM)

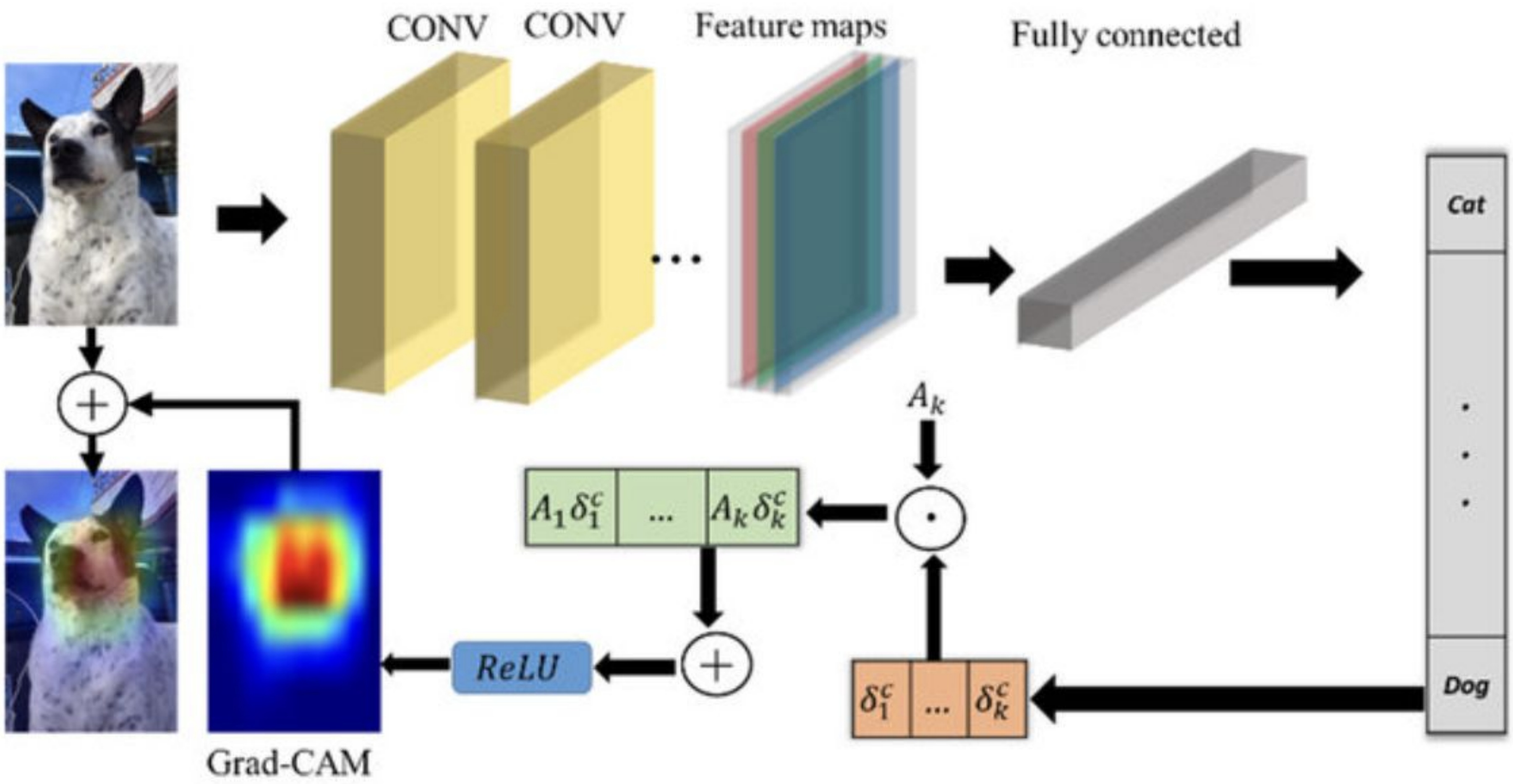
- A method for CNNs adopted in Computer Vision
  - even though it can be adapted to other domains, including time series
- Goal: understanding at which parts of an input a convolutional layer “looks” for a certain input
- Idea:
  - consider the gradients (backprop) on the ***last*** convolutional layer (i.e., the one before the dense layers)
  - analyze which regions are activated in the feature maps
  - generate a heatmap highlighting the parts of the input more relevant for classification

Every Ware Lab

GradCAM algorithm

- Forward-propagate the input image through the CNN.
- Obtain the raw score for the class of interest
  - (i.e., activation of the last neuron before applying softmax)
- Set all other class activations to zero.
- Back-propagate the gradient of the class of interest to the last convolutional layer before the fully connected layers
- Weight each feature map “pixel” by the gradient for the class.

GradCAM pipeline



Every Ware Lab

GradCAM formulas

To generate the heatmap:

$$\underbrace{ReLU}_{\text{Pick positive values}} \left(\sum_k \alpha_k^c A^k \right)$$

feature maps

$$\alpha_k^c = \overbrace{\frac{1}{Z} \sum_i \sum_j}^{\text{global average pooling}} \underbrace{\frac{\delta y^c}{\delta A_{ij}^k}}_{\text{gradients via backprop}}$$

Deep Explanation

Problems of Posthoc Explanations

- Approaches discussed so far are based on ***posthoc interpretability analysis***:
  - the network architecture is chosen first, and afterwards one aims to interpret the trained model or the learned high-level features
- Explanations significantly change based on the model used to generate explanation!
- Indeed, they come from a separate modeling process with strong priors, that are not part of training
- Also, posthoc methods sometimes create explanations that do not make sense to humans

Every Ware Lab

Deep Explanation

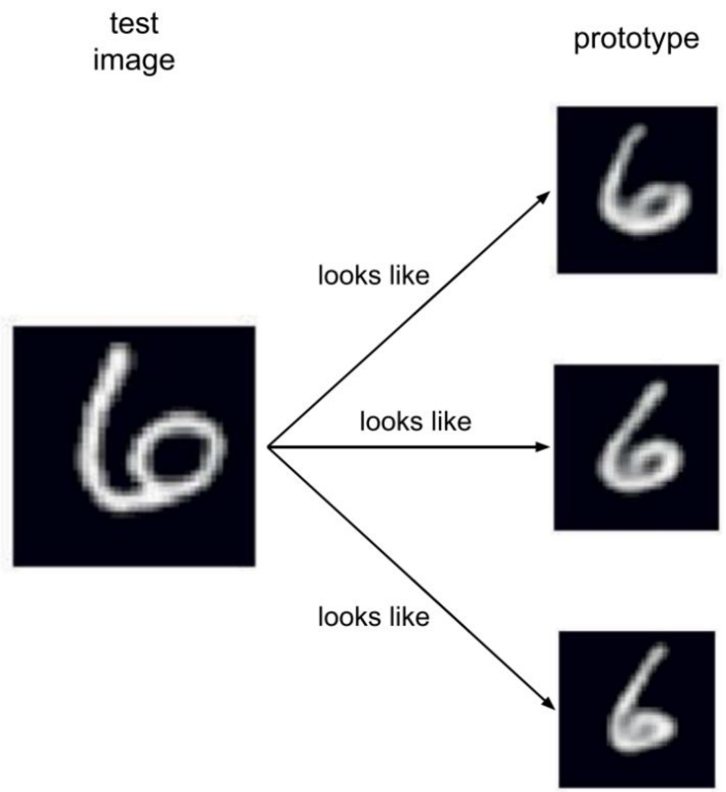
- The goal of *deep explanation* approaches is to create neural networks that are designed with interpretability in mind
- Hence, it is possible to precisely extract from the network the inner reasoning adopted for predictions
- In the following, we will consider one specific approach that is called *model prototypes*

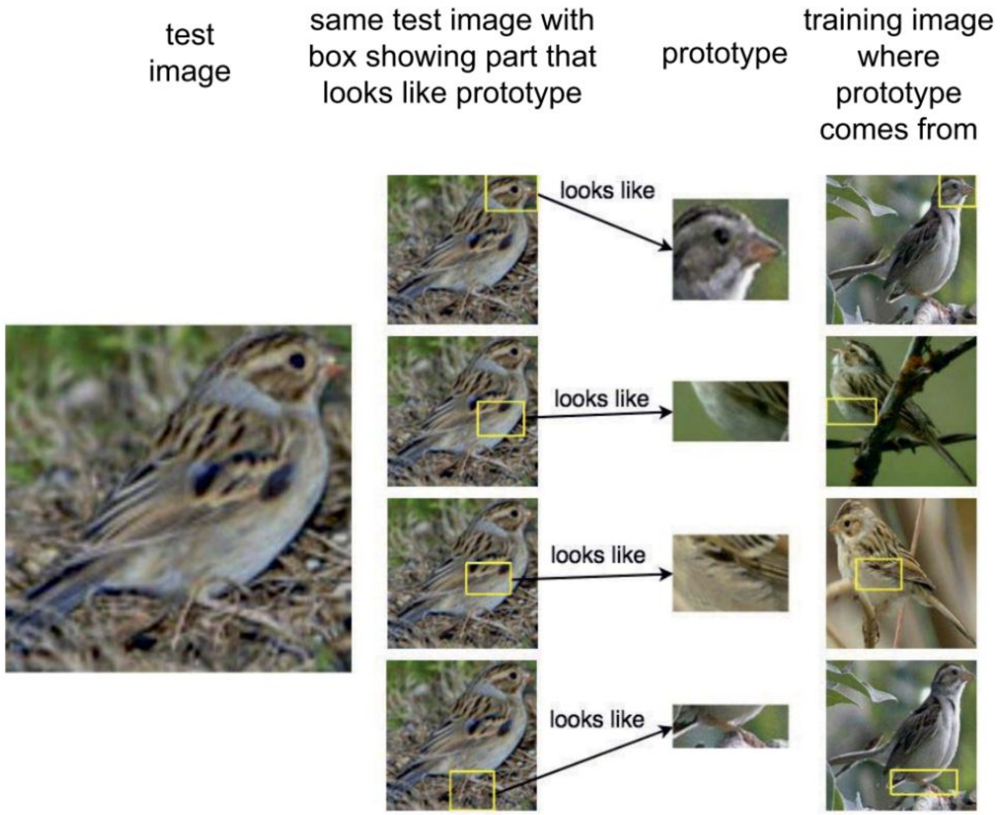
Prototype-based approaches

- Instead of explaining which parts of the input were important for classification…
- …*use-case based reasoning* to explain model its predictions based on similarity to prototypical cases
- A ***prototype*** is a data point that is very close or identical to an observation in the training set, and a limited set of prototypes should be representative of the whole data set.
- Specifically designed *prototype classifier:* inputs are classified based on their proximity to prototypes
  - note: prototypes may be extracted from the training set OR they may be learned by the model
    - we will see an approach that *learns* the prototypes

Every Ware Lab

Examples of explanations through prototypes





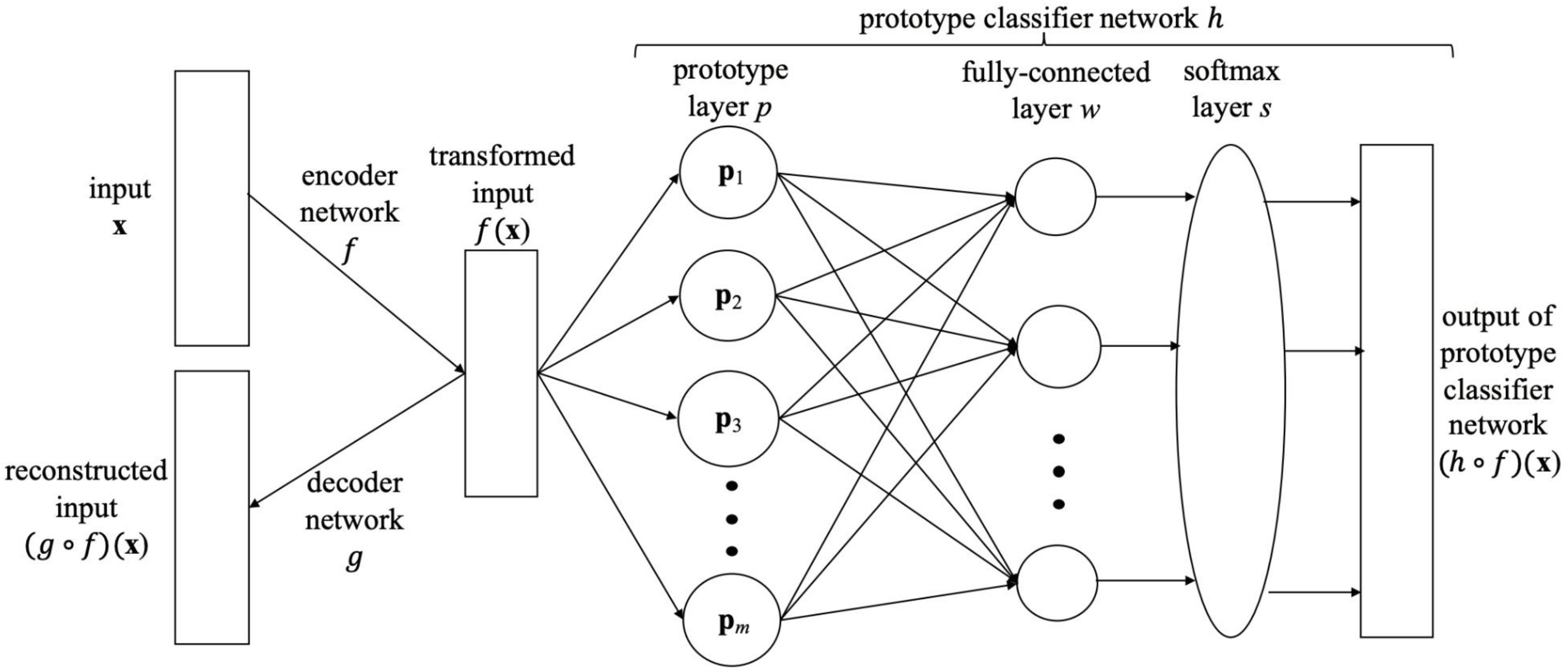
Rudin, C., Chen, C., Chen, Z., Huang, H., Semenova, L., & Zhong, C. (2022). Interpretable machine learning: Fundamental principles and 10 grand challenges. *Statistic Surveys*, *16*, 1-85.

Ware
Lab

An interpretable Deep Learning model based on *learning* Prototypes

Three main components:

- an autoencoder for to learn a latent space representation
- a prototype layer to learn *m* prototypes
- fully connected and softmax layers for classification



Li, O., Liu, H., Chen, C., & Rudin, C. (2018, April). Deep learning for case-based reasoning through prototypes: A neural network that explains its predictions. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 32, No. 1).

Every Ware Lab

Prototypes

- The network learns ***m*** prototypes
  - ***m*** is a parameter that should be tuned empirically
  - typically it depends on the specific task (e.g., number of classes, variety of patterns to capture, etcetera)
- Given an input data point ***z*** in the latent space, the squared L^2 distance is used to compute the distance between ***z*** and each prototype:

$$p(\mathbf{z}) = \left[\|\mathbf{z} - \mathbf{p}_1\|_2^2, \quad \|\mathbf{z} - \mathbf{p}_2\|_2^2, \quad \ldots \quad \|\mathbf{z} - \mathbf{p}_m\|_2^2 \right]^\top$$

- The explanation is usually provided with the *k* prototypes more similar to the input

Every Ware Lab

The learning objective

- The network has to learn **in parallel**
  - the latent feature space (using the autoencoder)
  - the classification task
  - the prototypes

Reconstruction loss

$$R(g \circ f, D) = \frac{1}{n} \sum_{i=1}^{n} \|(g \circ f)(\mathbf{x}_i) - \mathbf{x}_i\|_2^2.$$

Classification loss (cross-entropy)

$$E(h \circ f, D) = \frac{1}{n} \sum_{i=1}^{n} \sum_{k=1}^{K} -\mathbb{1}[y_i = k] \log((h \circ f)_k(\mathbf{x}_i))$$

Every Ware Lab

Prototype regularization terms

- Two interpretability regularization terms to learn the prototypes
- The first requires each prototype to be as close as possible to at least one of the training examples in the latent space:

$$R_1(\mathbf{p}_1, ..., \mathbf{p}_m, D) = \frac{1}{m} \sum_{j=1}^{m} \min_{i \in [1,n]} \|\mathbf{p}_j - f(\mathbf{x}_i)\|_2^2$$

- The second requires every training example to be as close as possible to one of the prototypes
  - (i.e., clustering the training examples around prototypes in the latent space)

$$R_2(\mathbf{p}_1, ..., \mathbf{p}_m, D) = \frac{1}{n} \sum_{i=1}^{n} \min_{j \in [1,m]} \|f(\mathbf{x}_i) - \mathbf{p}_j\|_2^2.$$

Every Ware Lab

Putting all together: the loss function

$$L((f,g,h),D) = E(h \circ f, D) + \lambda R(g \circ f, D) \\ + \lambda_1 R_1(\mathbf{p}_1, ..., \mathbf{p}_m, D) \\ + \lambda_2 R_2(\mathbf{p}_1, ..., \mathbf{p}_m, D),$$

Every
Ware
Lab

How to visualize prototypes?

- Prototypes are learned in the *latent* space
- To "visualize" them and provide the ones closest to the input as explanations, it is necessary to use the ***Decoder*** that can reconstruct them

Examples of learned prototypes from the MNIST dataset, in the pixel-space

Every Ware Lab

How to evaluate explanations?

How to understand if explanations are effective?

- Intuitively, it is possible to think that explanations that are ***convincing*** for the end users are also effective
- However, several other measures should be considered when building XAI models
- For instance, even though explanations may be convincing, they may be associated to mistakes of the underlying model
  - a challenge is indeed ***over-reliance***: the user trusts the system too much just because it provides explanations

Every Ware Lab

Measures of Explanation Effectiveness

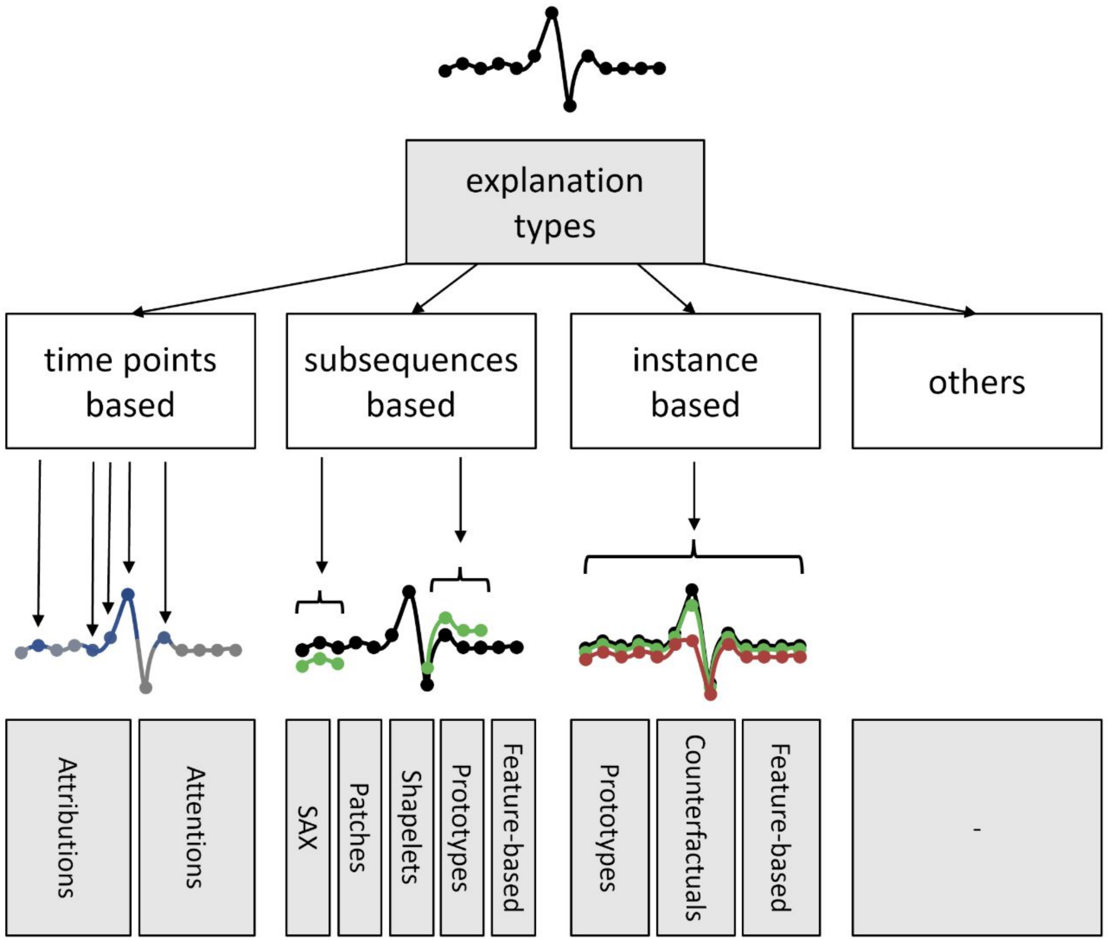
- User Satisfaction
  - clarity/utility of the explanation
  - usually assessed through questionnaires
- Mental model
  - understanding individual decisions and/or the overall model
  - a metric that assesses how a user understands the underlying model
  - assessed by asking users to describe the explanation process

Measures of Explanation Effectiveness (cont.)

- Trust Assessment
  - trust is the cognitive factor influencing the system's perception
  - trust may be positive or negative!
  - usually assessed by questionnaires before and after using the XAI model
- Correctability
  - assessing if XAI approaches are capable of identifying errors
  - error detection may lead to correction and continuous training
  - usually assessed with automatic methods

XAI and Time Series

Explaining Time Series



Theissler, A., Spinnato, F., Schlegel, U., & Guidotti, R. (2022). Explainable AI for time series classification: a review, taxonomy and research directions. *IEEE Access*, *10*, 100700-100724.

Every Ware Lab

Time-points based explanations

- Assign a relevance score to every time point of a time series
  - i.e., how much each sample contributes to the model's decision
- **Attributions methods**
  - using external methods (e.g., LIME, GradCam)
- **Attentions methods**
  - use an internal mechanism to incorporate a special focus, i.e., attention, of the network onto certain parts of the input

Every Ware Lab

Subsequence based explanations

- **Goal:** identifying sub-parts of a time series responsible for the model's outcomes

- **How?** extracting motifs (i.e., repeating patterns) from the times series and training interpretable models on them

Instance based explanations

- Using the whole temporal window to express the reasons for the classification
- Approaches:
  - **Feature-based explanations:** extracting higher level features and using standard methods to derive the most important features for classification
  - **Prototype-based explanations:** using real records sampled from the dataset that are important and meaningful because they are similar to the input, or learned ones (as in the Model Prototypes method)
  - **Counterfactual-based explanations:** showing the minimal changes in the input time window leading to a different decision outcome. For time series, minimality typically refers to a notion of distance between time series! It requires perturbation methods that are specific for time series.

Every Ware Lab

Ambient Intelligence Use Case 1: *Explainable Smart-Home Human Activity Recognition*

Every
Ware
Lab

Applying XAI to HAR

- In order to understand how XAI can be applied in Ambient Intelligence, we will see an application in HAR
- In particular, the first use case is based on this paper:

Arrotta, L., Civitarese, G., & Bettini, C. (2022). ***Dexar: deep explainable sensor-based activity recognition in smart-home environments.*** *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, *6*(1), 1-30.

Every Ware Lab

DeXAR

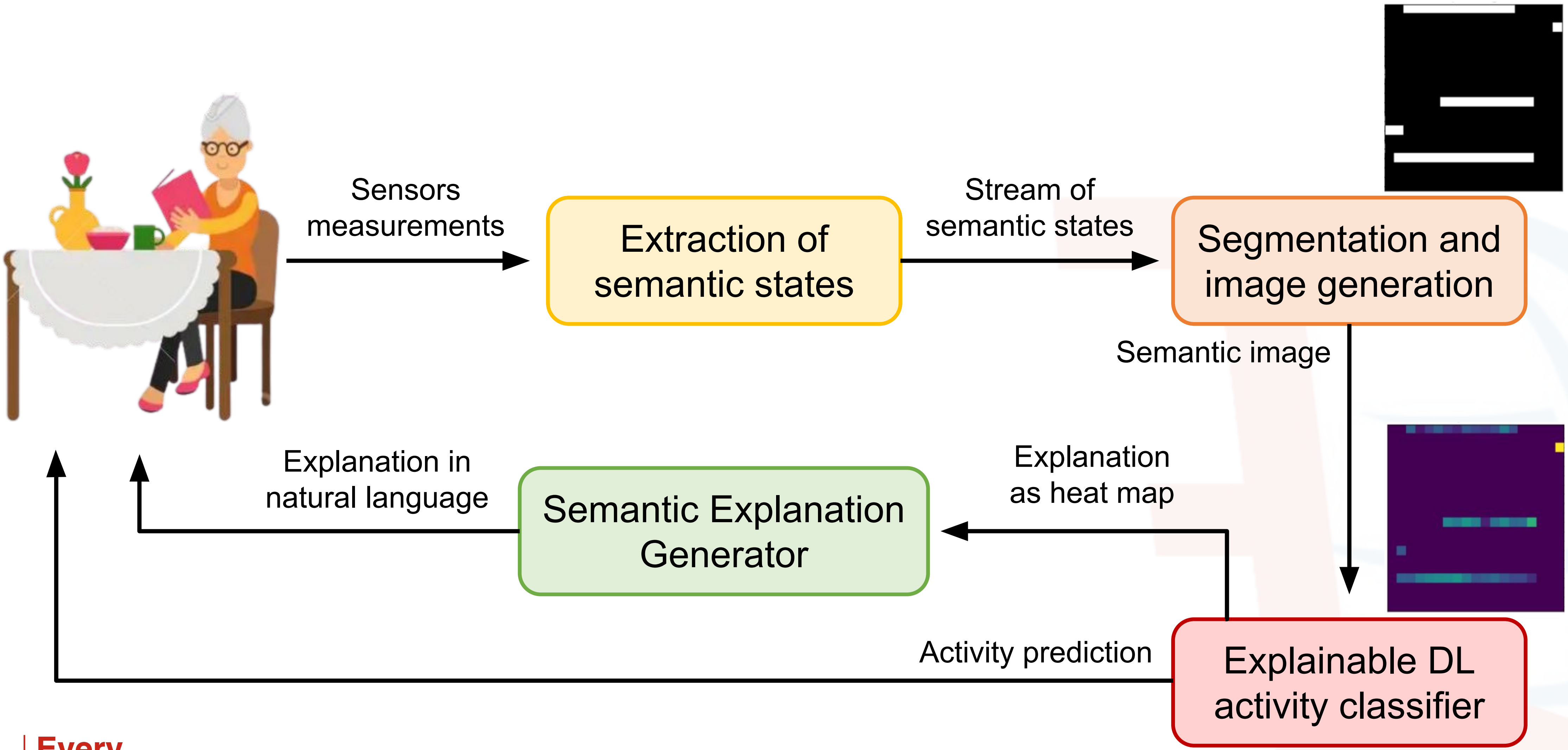
A methodology for explainable sensor-based ADLs recognition based on Deep Learning models.

Steps:

1. Transforms sensors' data into semantic images
2. Uses existing XAI methods for Computer Vision to obtain explanations as heat maps
3. Translates heat maps into natural language sentences

The activity is **EATING** mainly because Bob has just cooked a hot meal

DeXAR Data Flow



Every
Ware
Lab

Extraction of Semantic States

- XAI makes sense if we can understand the semantics of the explanation
- However, raw sensors' measurements are hard to explain to non-expert users
- For this reason, it is necessary to pre-process sensor data to obtain a stream of semantic states:
  - High-level information with a clear semantic describing what happened within a time interval

Extraction of Semantic States (Environmentals Sensors)

- Usually have a binary output: ON or OFF at specific time instants
- Environmental sensor events can be easily mapped to semantic states:
  - A pressure mat sensor *P* on the kitchen chair outputs the value ON at time *t1*
  - At time *t2*, *P* outputs the value OFF
  - DeXAR derives the semantic state *using_kitchen_chair [t1, t2]*

Every
Ware
Lab

Extraction of Semantic States (Wearables)

- Mapping raw inertial sensors' data to semantic states is not easy
  - e.g., an accelerometer that continuously generates values on three axes at a high frequency
- Existing works suggest that sequences of low-level physical activities can reveal higher-level activities
  - it's possible to use state-of-the-art models to derive simple low-level activities from inertial data
  - e.g., generating semantic state *sitting [t1, t2]* when a model analyzing smartwatch data infers the sitting activity from *t1* to *t2*

Every Ware Lab

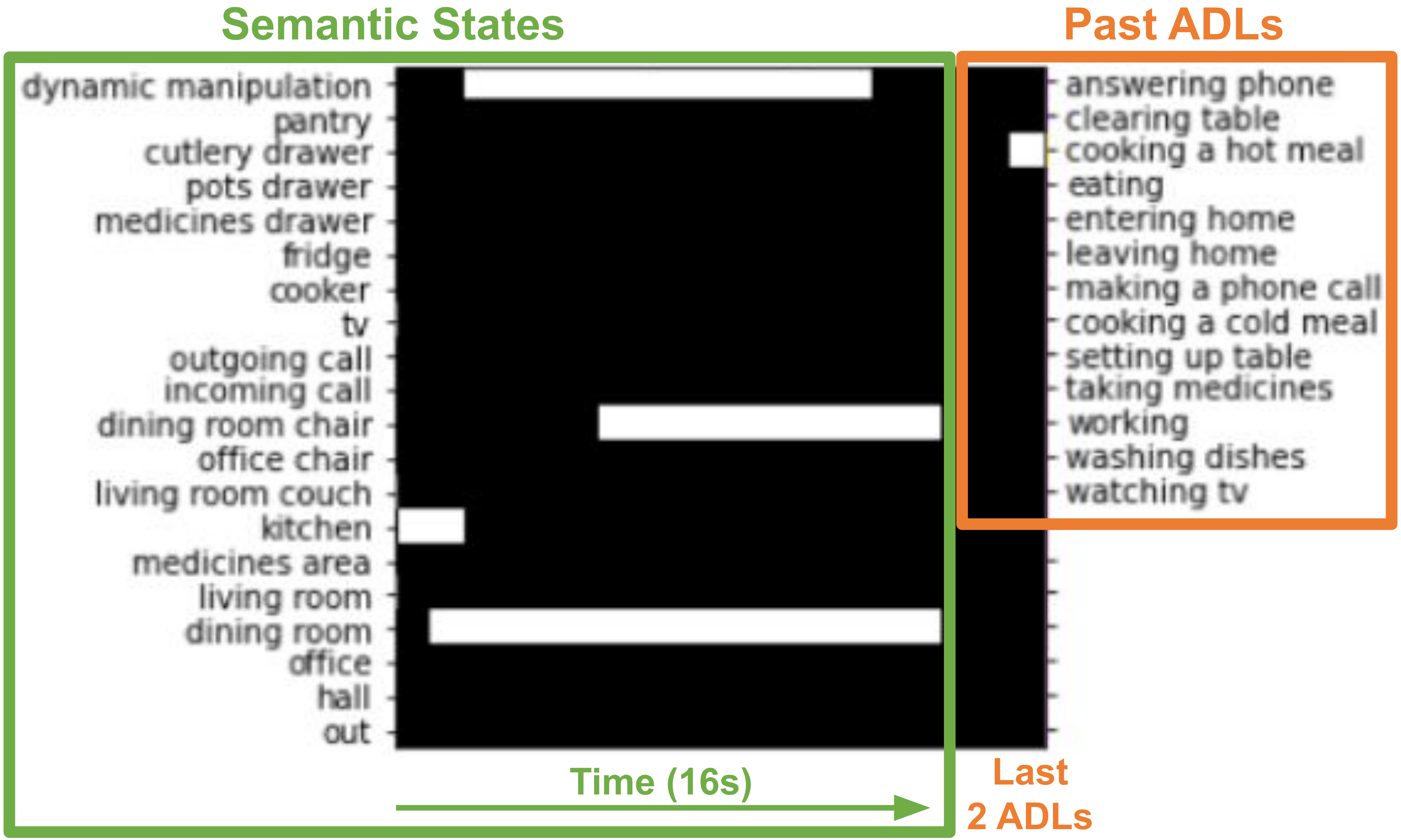
Semantic Image Generation

- For each segment of data, a semantic image is generated (i.e., the input of the ADLs classifier)

- A semantic image includes:
  - The semantic states observed within the segment
  - The most recent *K* activities that the resident performed before the current one

Every
Ware
Lab

Semantic Image Generation



XAI Explanation

- Given a semantic image it is possible to apply any XAI method for computer vision
  - e.g., LIME, GradCAM, …
- In the following we will see an application of the *prototypes* approach

Every
Ware
Lab

Explaining using Model Prototypes

Every
Ware
Lab

How to provide a single explanation from prototypes?

ALGORITHM 1: Generating a heatmap from prototypes

Input: The input image img, the m closest prototypes $P = \{p_1, p_2, \ldots, p_m\}$ to img, a threshold pt
Output: A heatmap h

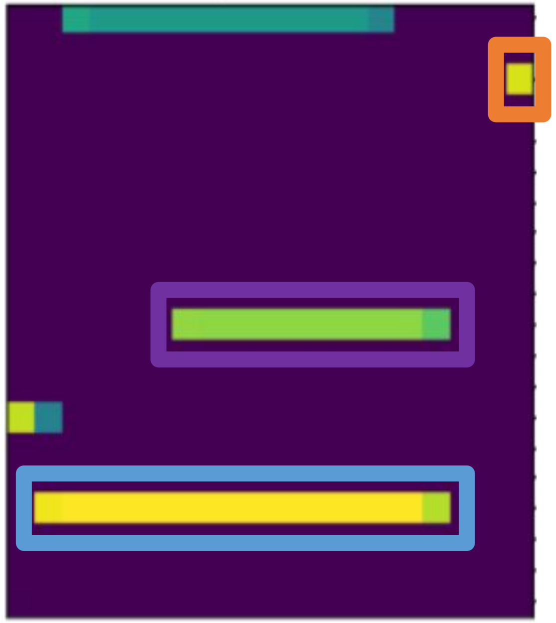
```
h ← empty heat map
for each row i of img do
    for each column j of img do
        if img[i, j] > 0 then
            τ ← 0
            for p ∈ P do
                if p[i, j] > pt then
                    τ ← τ + 1
                end
            end
            h[i, j] ← τ/m
        end
    end
end
return h
```

Every Ware Lab

Semantic Explanation Generator

The heat map may be useful for data scientists, but not for end users. Since each pixel is associated with a semantic, it is possible to build an explanation in natural language:

1. Extracting the most relevant features from the heat map (using a threshold)
2. Using these features to generate a sentence (e.g., with heuristics or using an LLM)

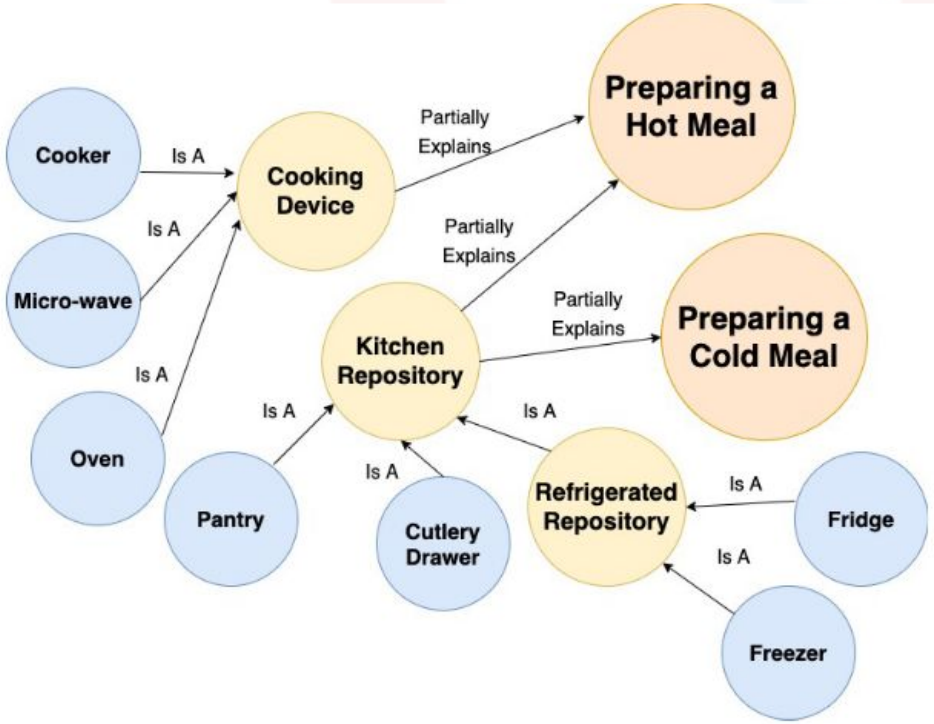


The activity is **EATING** mainly because Alice has been in the **dining room** while sitting on a **dining room table chair**. Moreover, she has just **cooked a hot meal**

Every Ware Lab

Explanation Score

- A metric to ***automatically*** evaluate the consistency of an explanation with common-sense knowledge about the HAR domain
  - e.g., washing dishes is typically performed in the kitchen, after eating and clearing the table
- "The feature *f* was important to classify *A* according to XAI. Does this make sense considering common-sense knowledge?"
- The common-sense knowledge is encoded through a semantic model that defines which features ***partially explain*** each activity



Every Ware Lab

How to compute the Explanation Score

F^* = Semantic States that were important in the heatmap for an image *img* (based on a threshold)

rel(f,img) = relevance of a feature *f* (intensity of pixel)

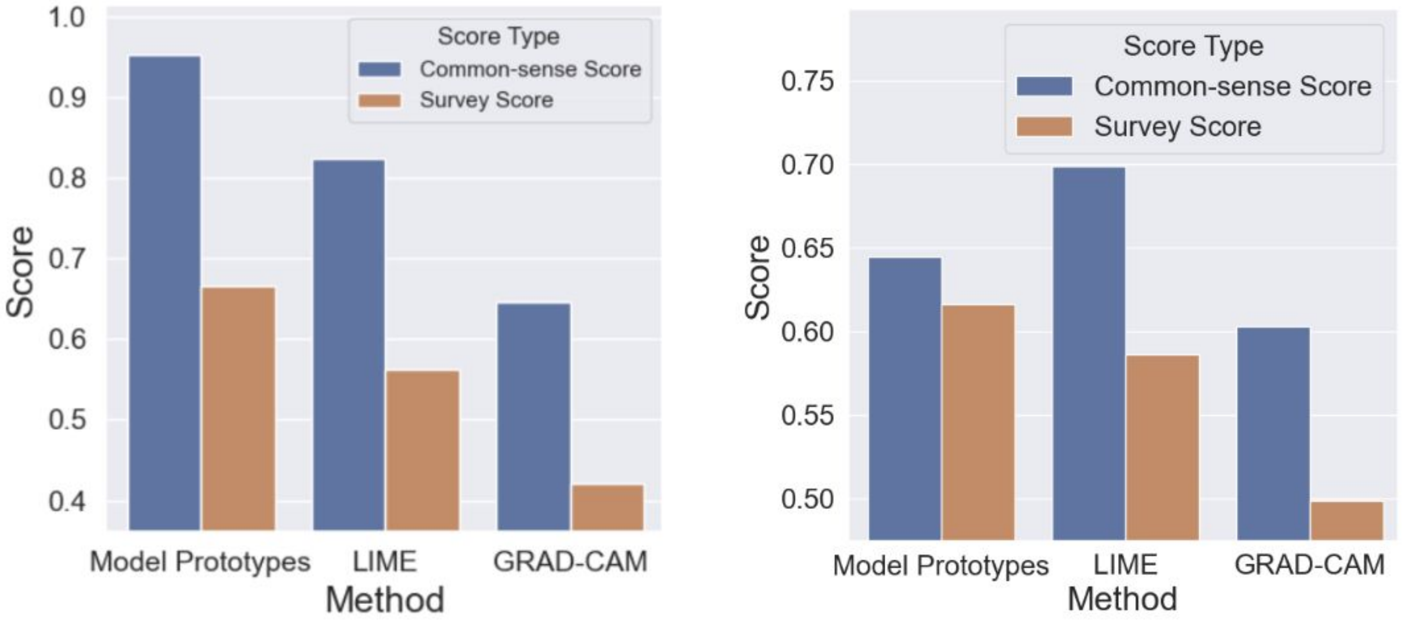
cr(f,A) = common-sense relevance of a feature *f* in partially explaining an activity *A*

$$cr(f,A) = \begin{cases} rel(f,img) & \text{if } f \text{ partially explains } A \\ -rel(f,img) & \text{otherwise} \end{cases}$$

$$ExplanationScore(F^\star, A) = \begin{cases} \dfrac{\sum_{f \in F^\star} cr(f,A)}{\sum_{f \in F^\star} |cr(f,A)|} & \text{if } F^\star \neq \emptyset \\ -1 & \text{otherwise} \end{cases}$$

Every Ware Lab

Explanation Score vs User-Based score

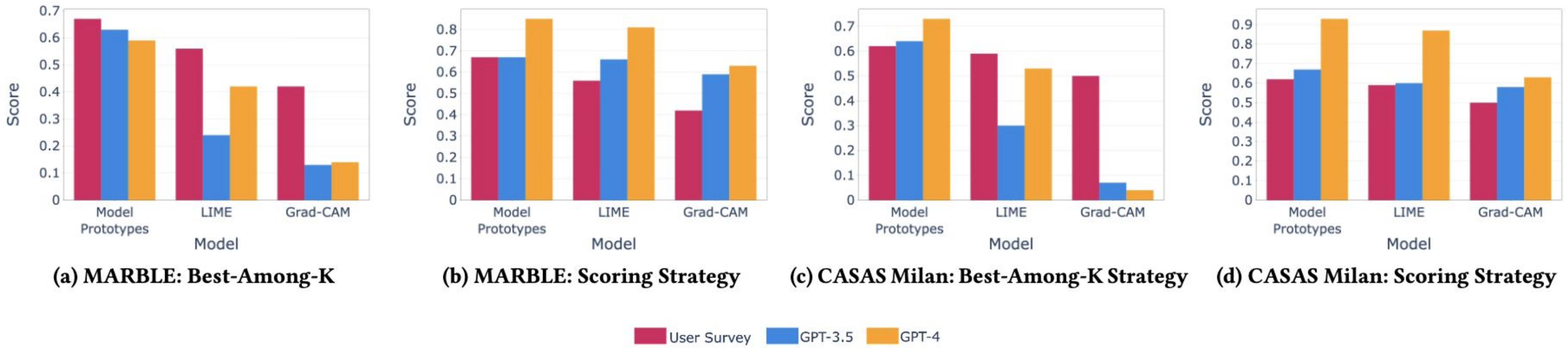


(a) Our dataset

(b) CASAS

Every
Ware
Lab

“LLM-as-a-judge” may be aligned with users!



(a) MARBLE: Best-Among-K **(b) MARBLE: Scoring Strategy** **(c) CASAS Milan: Best-Among-K Strategy** **(d) CASAS Milan: Scoring Strategy**

Ambient Intelligence Use Case 2: *Explainable Nursing Activity Recognition*

Every
Ware
Lab

Nursing Care Activities

- In healthcare scenarios, monitoring patients' activities is not the only important task
- Detecting activities of healthcare professionals (e.g., nurses in this case) may enable several applications
  - e.g., automatic record activities to reduce documentation time, checking compliance with care routines for a given patient, identification of risky activities (e.g., not washing hands before taking blood samples).
- This use case is based on the following paper:

Jeyakumar, J. V., Sarker, A., Garcia, L. A., & Srivastava, M. (2023). ***X-CHAR: A Concept-based Explainable Complex Human Activity Recognition Model.*** *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, *7*(1), 1-28.

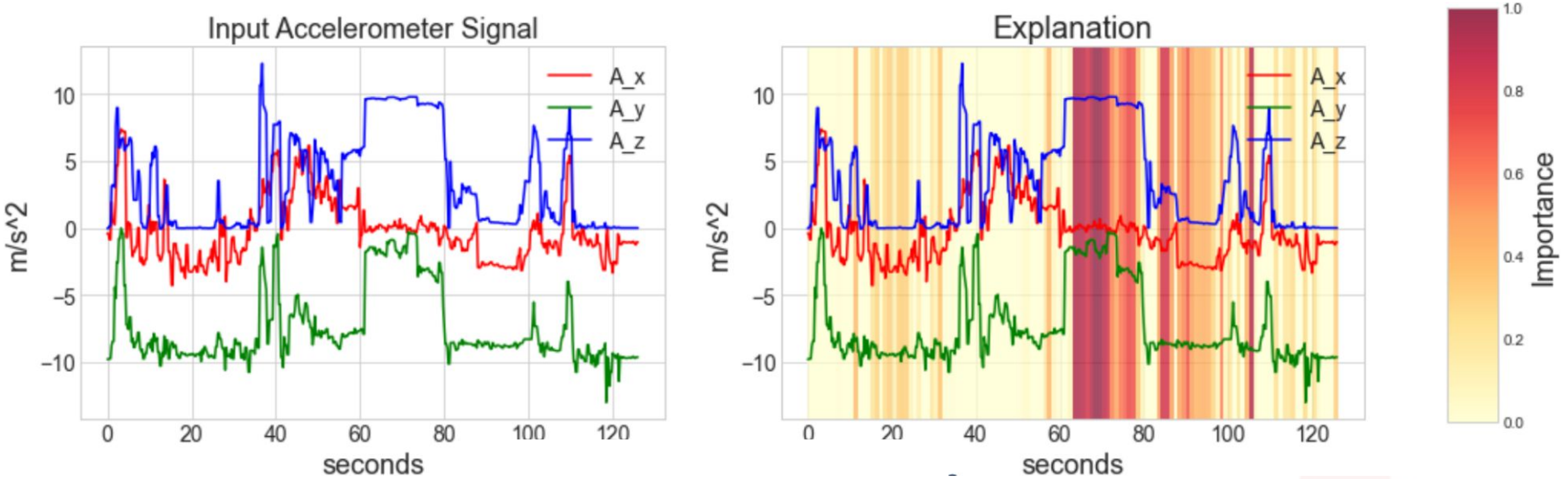
Every Ware Lab

The problem of explaining time-series data with continuous values

- In the previous use case showed a XAI method where raw sensor data could be associated with a semantic
  - This association made it possible to generate an explanation for end-users
- Now, let’s assume that nurses only have a wearable device (e.g., a smartphone) generating continuously inertial sensor data
- This setting is more challenging
  - directly applying XAI approaches may only be useful to highlight the portion of time series that was important for the classifier

Every Ware Lab

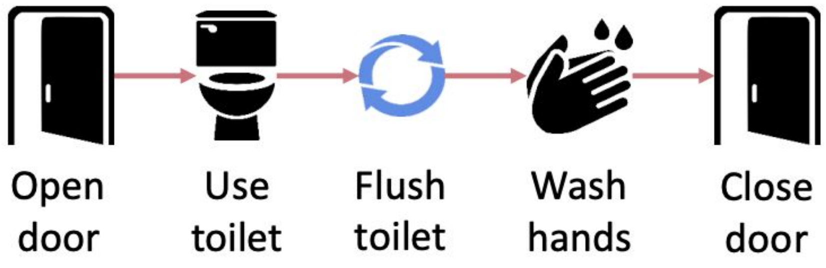
Example: GradCAM applied to accelerometer data



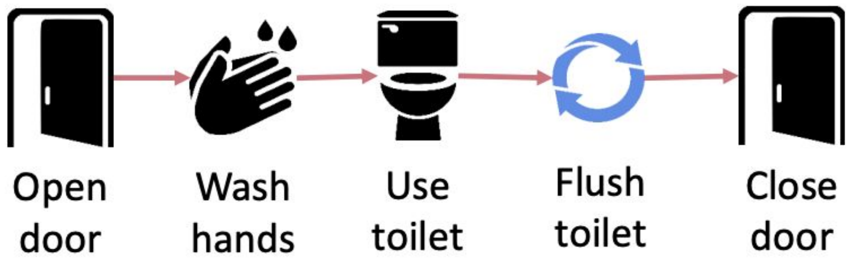
Every Ware Lab

Simple and Complex Activities

- As also mentioned in the previous use case, complex activities may be recognized as sequences of simpler low-level activities
- Hence, it may be possible to explain complex activities based on the sequence of detected simpler activities
- From now on, we will refer to a simple activity as "concept"



Using Restroom (Hygienic)



Using Restroom (Unhygienic)

Every Ware Lab

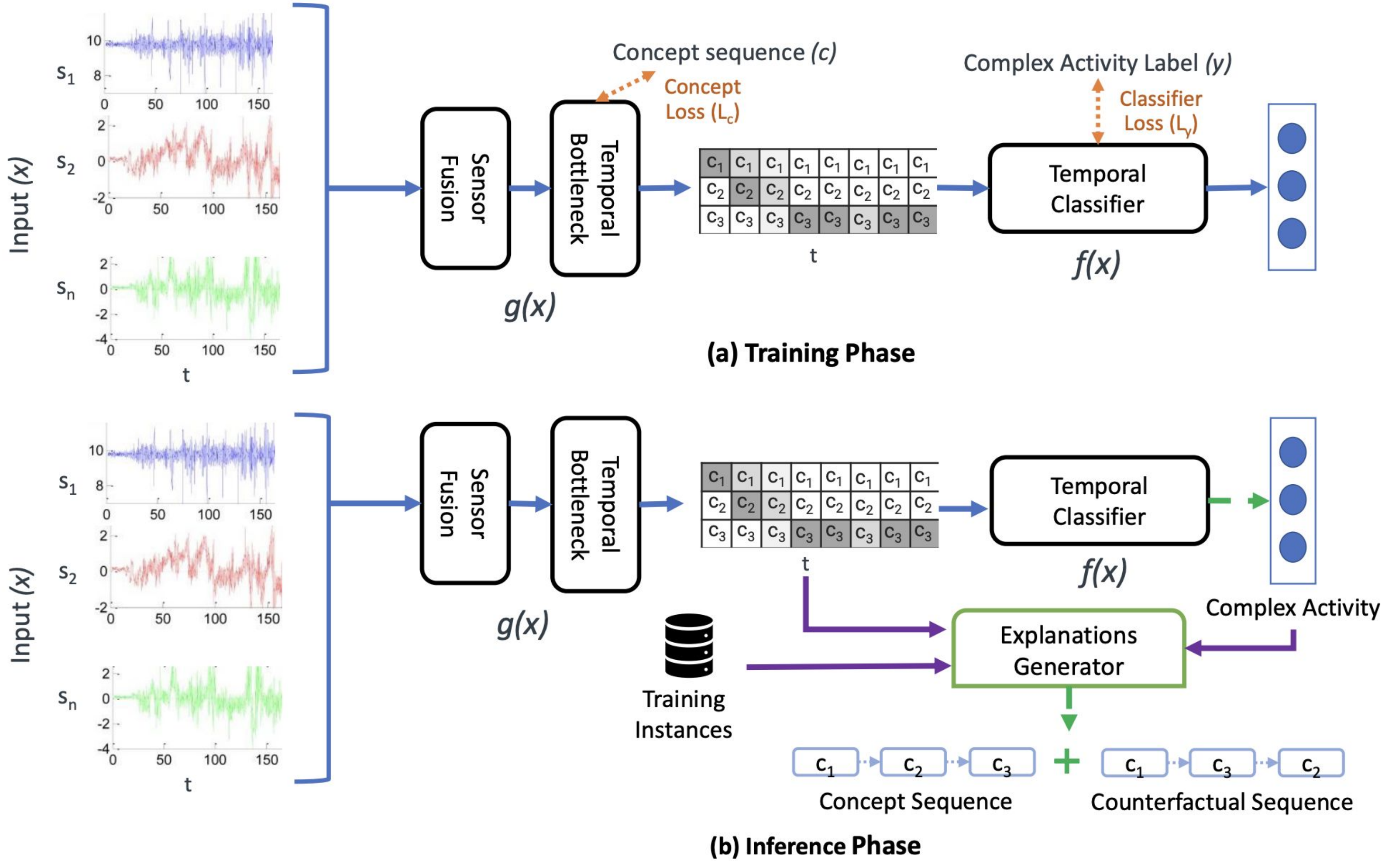
Activities and concepts in Nurse Care

Complex Nurse Activity	Concept Sequence	Concepts
Physiological Measurement	Vitals → Blood collection → Blood Glucose	check vitals, collect blood, check blood glucose, oral care, clean patient, drips
	Blood collection → Vitals → Blood glucose	
Patient cleaning	Vitals → Oral care → Clean	
	Oral care → Vitals → Clean	
	Oral care → Blood glucose → Clean	
	Blood glucose → Oral care → Clean	
Unsanitary Operations	Clean → Blood glucose → Vitals	
	Clean → Oral care → Vitals	
	Vitals → Clean → Oral care	
	Oral care → Clean → Blood glucose	
Safe IV/Drips Procedure	Vitals → Blood collection → Drips → Vitals	
	Vitals → Drips → Vitals	
	Vitals → Drips → Blood collection → Vitals	
Unsafe IV/Drips Procedure	Vitals → Blood collection → Drips	
	Blood collection → Drips → Vitals	
	Drips → Blood collection → Vitals	

X-CHAR:
eXplainable Complex Human Activity Recognition

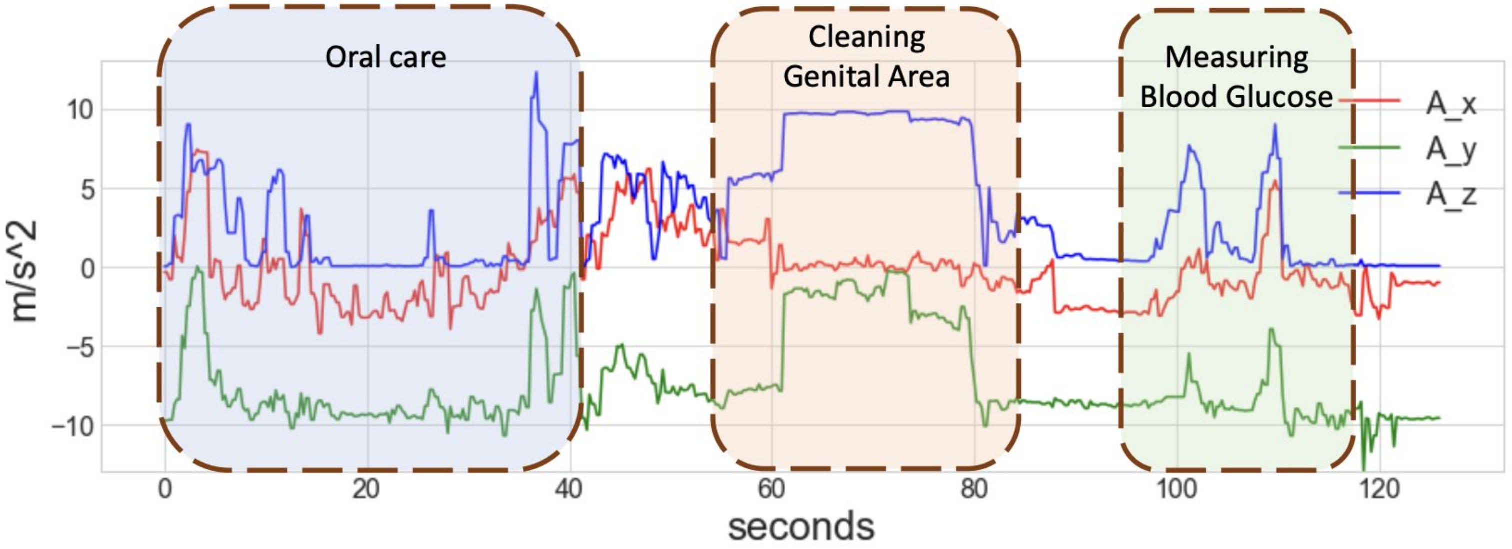
- The model learns to recognize concepts, and then complex activities based on sequences of concepts (end-to-end)

- During classification, the model provides a ***counterfactual explanation*** to explain the detected complex activity!



Every Ware Lab

Example of a Counterfactual Explanation



Explanation: Classified as ***'Unsanitary Care'*** because the model says the nurse performed the following sequence of simple activities:

Oral care -> Cleaning Genital area -> Measure Blood Glucose

Counterfactual Explanation: It would be ***'Cleaning the Patient'*** activity had the sequence been

Measure Blood Glucose -> Oral care -> Cleaning Genital Area

Every Ware Lab

How X-CHAR generates a counterfactual explanation

- Storing a set ***C*** of all the concept sequences generated by the model on the training set data
- Given a test input data *x*, obtain the corresponding concept sequence *c*
- Find the nearest c_{ex} in ***C*** such that has the minimum distance with *c* but classified with a different class
- c_{ex} is the counterfactual explanation!

Every Ware Lab

References

[1] Molnar, C. (2020). ***Interpretable machine learning***. https://christophm.github.io/interpretable-ml-book/index.html

[2] Li, O., Liu, H., Chen, C., & Rudin, C. (2018, April). ***Deep learning for case-based reasoning through prototypes: A neural network that explains its predictions.*** In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 32, No. 1).

[3] Arrotta, L., Civitarese, G., & Bettini, C. (2022). ***Dexar: deep explainable sensor-based activity recognition in smart-home environments.*** Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 6(1), 1-30.

Every Ware Lab